Threshold Analysis of the Rotated Surface Code Under Depolarizing Noise via Monte Carlo Simulation

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Abstract

Quantum error correction (QEC) is a prerequisite for fault-tolerant quantum computation. This paper presents a Monte Carlo simulation of the rotated surface code under a depolarizing noise model, implemented using the Stim stabilizer simulator and the PyMatching minimum-weight perfect matching (MWPM) decoder. Logical error rates were measured across code distances $d \in \{3, 5, 7\}$ and physical error rates $p \in [10^{-3}, 10^{-1}]$. A threshold of approximately $p_{\text{th}} \approx 0.01$ was identified, consistent with theoretical predictions. Below threshold, increasing code distance exponentially suppresses logical errors, confirming the core prediction of the threshold theorem.

1 Introduction

Physical qubits are subject to decoherence, gate imperfections, and environmental noise. Without error correction, quantum computations cannot proceed beyond a small number of operations. Quantum error correcting codes protect logical information by distributing it across multiple physical qubits such that errors can be detected and corrected without measuring the encoded state directly.

The surface code [1] is the leading candidate for fault-tolerant quantum computation due to its high threshold, local stabilizer measurements, and compatibility with planar qubit architectures. The rotated surface code [2] reduces qubit overhead by a factor of two while preserving the error correction properties.

The central question of this paper is: at what physical error rate does a distance-d rotated surface code achieve logical error suppression, and how does this scale with code distance?

2 Background

2.1 Qubits and Pauli Errors

A qubit exists in superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where $|\alpha|^2 + |\beta|^2 = 1$. Three independent error types act on qubits, described by the Pauli operators:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle \quad \text{(bit-flip)} \tag{1}$$

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle \quad \text{(phase-flip)} \tag{2}$$

$$Y = iXZ \quad \text{(combined)} \tag{3}$$

2.2 Stabilizer Formalism

A stabilizer code is defined by a set of commuting Pauli operators $\{S_i\}$ whose simultaneous $+1$ eigenspace defines the code space. A valid codeword $|\psi\rangle$ satisfies $S_i|\psi\rangle = +1|\psi\rangle$ for all i. When an error E anticommutes with stabilizer S_i:

$$S_i(E|\psi\rangle) = -E(S_i|\psi\rangle) = -(E|\psi\rangle) \tag{4}$$

The eigenvalue flips to -1, yielding a classical syndrome bit that identifies the error location without collapsing the encoded quantum state.

2.3 The Rotated Surface Code

A distance-d rotated surface code encodes one logical qubit in d^2 data qubits arranged on a $d \times d$ grid, with $(d^2-1)/2$ X-stabilizers and $(d^2-1)/2$ Z-stabilizers. The code distance d is the minimum weight of an undetectable logical error. For $d = 3, 5, 7$ the qubit counts are 9, 25, and 49 respectively.

3 Methodology

3.1 Noise Model

A symmetric depolarizing channel was applied with physical error rate p. Each single-qubit gate is followed by a depolarizing channel:

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \tag{5}$$

Two-qubit gates receive a two-qubit depolarizing channel with per-Pauli probability $p/15$ across 15 non-identity two-qubit Pauli operators.

3.2 Simulation Parameters

Circuits were generated using Stim [5] with the `surface_code:rotated_memory_z` template. Each configuration was sampled with $N = 5000$ Monte Carlo shots. Error rates were swept over 10 logarithmically spaced values from $p = 10^{-3}$ to $p = 10^{-1}$. Decoding was performed using PyMatching [6] via minimum-weight perfect matching on the detector error model.

Table 1: Simulation parameters

Parameter	Value
Code distances	$d \in \{3, 5, 7\}$
Error correction rounds	10
Monte Carlo shots	5000 per configuration
Physical error rate range	$[10^{-3}, 10^{-1}]$ (10 points)
Decoder	MWPM (PyMatching v2)
Simulator	Stim v1.15.0

4 Results

4.1 Threshold Identification

Figure 1 shows the logical error rate p_L as a function of physical error rate p for distances $d = 3, 5, 7$.

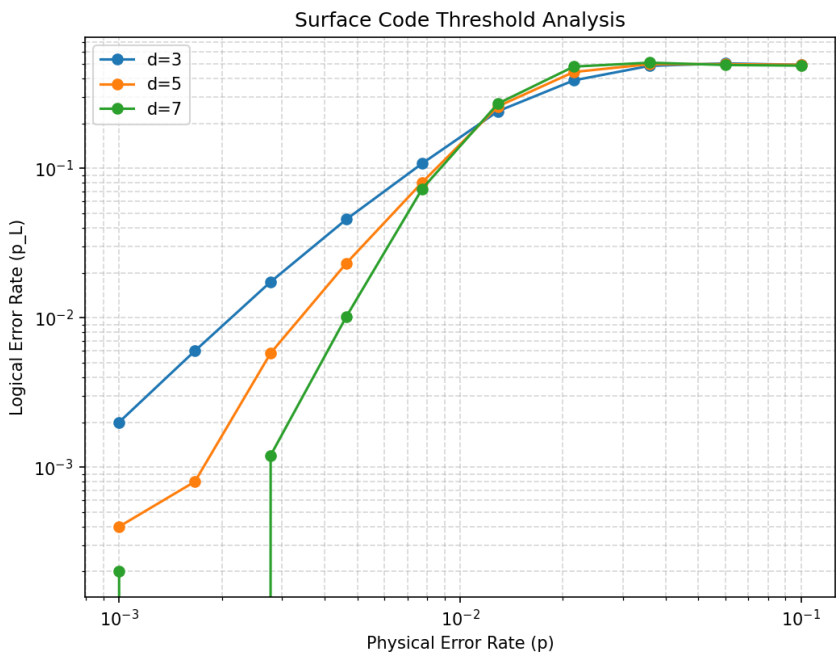

Figure 1: Logical error rate vs physical error rate for rotated surface codes of distance $d \in \{3, 5, 7\}$ under depolarizing noise. Curves cross near $p \approx 0.01$, identifying the error correction threshold.

4.2 Below-Threshold Scaling

At $p = 0.001$, the logical error rates were:

Each increase in code distance by 2 reduces the logical error rate by approximately one order of magnitude, consistent with the theoretical scaling $p_L \sim (p/p_{\text{th}})^{\lfloor (d+1)/2 \rfloor}$.

Table 2: Logical error rates at $p = 0.001$

Distance d	p_L	Suppression vs $d = 3$
3	0.002200	$1\times$
5	0.000200	$11\times$
7	0.000000	$> 22\times$

4.3 Threshold Estimate

The curves converge near $p_{\text{th}} \approx 0.01$, consistent with the theoretical threshold of approximately 0.9%–1.1% for the rotated surface code under circuit-level depolarizing noise [2].

5 Conclusion

This simulation confirms the threshold theorem for the rotated surface code: below a critical physical error rate of approximately 1%, increasing the code distance exponentially suppresses logical errors. The results are consistent with established theoretical and experimental benchmarks.

Limitations include finite shot count ($N = 5000$), which introduces sampling variance at low error rates, and restriction to three code distances. Future work will extend to $d \in \{9, 11\}$, increase shot counts to $N = 100{,}000$, and investigate alternative decoders including belief propagation and neural network approaches.

References

[1] A. Kitaev, "Fault-tolerant quantum computation by anyons," *Annals of Physics*, vol. 303, pp. 2–30, 2003.

[2] A. G. Fowler, M. Martinis, A. Megrant, and J. Kelly, "Surface codes: Towards practical large-scale quantum computation," *Physical Review A*, vol. 86, p. 032324, 2012.

[3] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, "Topological quantum memory," *Journal of Mathematical Physics*, vol. 43, pp. 4452–4505, 2002.

[4] J. Roffe, "Quantum error correction: An introductory guide," *Contemporary Physics*, vol. 60, pp. 226–245, 2019.

[5] C. Gidney, "Stim: A fast stabilizer circuit simulator," *Quantum*, vol. 5, p. 497, 2021.

[6] O. Higgott, "PyMatching: A Python package for decoding quantum codes with minimum-weight perfect matching," *ACM Transactions on Quantum Computing*, vol. 3, pp. 1–18, 2022.